

mosaic-go G5 P3/P3H

Evaluation kit with mosaic-G5 P3 or mosaic-G5 P3H high-accuracy receiver



Septentrio mosaic-go G5 is an evaluation kit integrating the mosaic-G5 P3™ or mosaic-G5 P3H™ high precision multi-frequency GNSS receiver module. It has been designed for rapid prototyping and evaluation of these modules. The mosaic-G5 P3 and P3H receivers bring high-performance and positioning availability to commercial drones, robots and other industrial mass market application requiring reliable high-accuracy positioning. The mosaic-G5 P3H module delivers sub-degree GNSS heading, in addition to accurate positioning.

KEY FEATURES

- ▶ **UART, USB-C, PPS and SD card slot**
- ▶ **Easy evaluation of rover functionality**
- ▶ **All-in-view satellite tracking: multi-constellation, quad-band GNSS receiver and triple-band heading receiver**
- ▶ **Best-in-class RTK centimeter-level positioning**
- ▶ **Dual antenna heading with mosaic-G5 P3H™**
- ▶ **Direct connection to popular autopilots**

BENEFITS

Performance excellence

The mosaic-G5 P3 offers full raw data with positioning measurements. Both P3 and P3H offer Galileo HAS positioning service compatibility¹ and are ideal for drones, robots, autonomous systems and space-constrained devices which require accurate positioning with a high degree of reliability.

Designed for automated assembly

The mosaic-G5 P3™ and mosaic-G5 P3H™ modules come with a comprehensive set of interfaces, commands, and data messages, which are fully documented. The RxTools software suite allows convenient receiver configuration and analysis.

Advanced technologies inside

Septentrio's **GNSS+** technologies enable accuracy and reliability under the toughest conditions. They include:

- ▶ **LOCK+** for robust tracking during high vibrations and shocks.
- ▶ **APME+** multipath mitigation to disentangle direct signal and those reflected from nearby structures.
- ▶ **AIM+ Advanced** functionality with jamming and spoofing detection and automatic mitigation.
- ▶ **IONO+** protection against ionospheric disturbances.

FEATURES

GNSS technology

789 hardware channels for simultaneous tracking of all visible supported satellite signals:

- ▶ GPS: L1C/A, L1C, L2C, L2PY, L5
- ▶ GLONASS: L1CA, L2CA, L2P, L3 CDMA
- ▶ Beidou: B1I, B1C, B2a, B2I, B2b, B3I
- ▶ Galileo: E1, E5a, E5b, E6
- ▶ QZSS: L1C/A, L1 C/B, L2C, L5, L6⁸

GNSS Heading (P3H only)

Galileo High Accuracy Service (HAS)¹

Galileo OSNMA

Full raw measurement data (P3 only)

5 constellations RTK (rover)

Septentrio's patented GNSS+ technologies

- ▶ **AIM+ Advanced** functionality with jamming and spoofing detection automatic interference mitigation
- ▶ **APME+** a posteriori multipath estimator for code and phase multipath mitigation
- ▶ **LOCK+** superior tracking robustness under heavy mechanical shocks or vibrations
- ▶ **IONO+** provides advanced protection against ionospheric disturbances.
- ▶ **RAIM+** receiver autonomous integrity monitoring

Protocols

Septentrio Binary Format (SBF)

NMEA 0183, v2.3, v3.03, V4.0

RTCM v3.x (MSM included) input

Interfaces

UART 1: 6-pin JST-GH (LVTTTL, up to 4 Mbps, direct autopilot connection)

UART 2: 10-pin 100 mil socket (LVTTTL, up to 4 Mbps, direct Bluetooth dongle connection)

USB C (2.0, HS up to 480Mbps)

1 GPIO user programmable

1 Configurable PPS out

SD Card for logging

1 Event marker

PERFORMANCE

RTK performance^{2,3,4}

Horizontal accuracy	0.6 cm + 0.5 ppm
Vertical accuracy	1 cm + 1 ppm
Initialization time	7 s

Other positioning modes accuracy^{2,3}

	Horizontal	Vertical
Standalone	1.2 m	1.9 m
DGNSS	0.4 m	0.7 m

Velocity accuracy

	3 cm/s
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GNSS attitude accuracy (P3H only)^{2,3}

	Heading	Pitch/Roll
Antenna Separation		
1m	0.15°	0.25°
5m	0.03°	0.05

Maximum update rate

Position	20 Hz
Measurements only (P3 only)	20 Hz

Latency⁵

	<10 ms
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Time precision

PPS resolution	1.4 ns
Event accuracy	< 3 ns

Time to first fix

Cold start ⁶	< 35 s
Warm start ⁷	< 10 s
Re-acquisition	1 s

Tracking performance (C/N0 threshold)

Tracking	20 dB-Hz
Acquisition	30 dB-Hz

PHYSICAL AND ENVIRONMENTAL

Package

Type	Casting metal enclosure
Size	74 x 44 x 11.4 mm
Weight	50 g

Electrical

Antenna preamplification range	15-50 dB SA 15-35 dB DA
Antenna bias voltage	3.0-5.5 V Built-in current limit (150 mA)

Input voltage USB or 5 VDC

Power consumption P3 0.59 W typ/0.77 Max

Dual Antenna (P3H only) 0.81 W typ/1.06 Max

Environmental

Operating temp -25 to 85° C
-13 to 185° F

Storage temp -55 to 85° C
-67 to 185° F

Certification CE, FCC, RoHS, WEEE, ISED



¹ Future-proof feature which will be roll out by regular software updates

² Open sky conditions

³ RMS levels

⁴ Baseline <40 km

⁵ 99.9%

⁶ No information available (no almanac, no approx. position)

⁷ Ephemeris and approx. position known

⁸ Will be rolled out via software updates

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G5 P3



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